

Stylized Fluid Shader: Documentation

Asset Store Package Version: 1.0.0

Target Pipeline: Universal Render Pipeline (URP) / Unity 6.3+

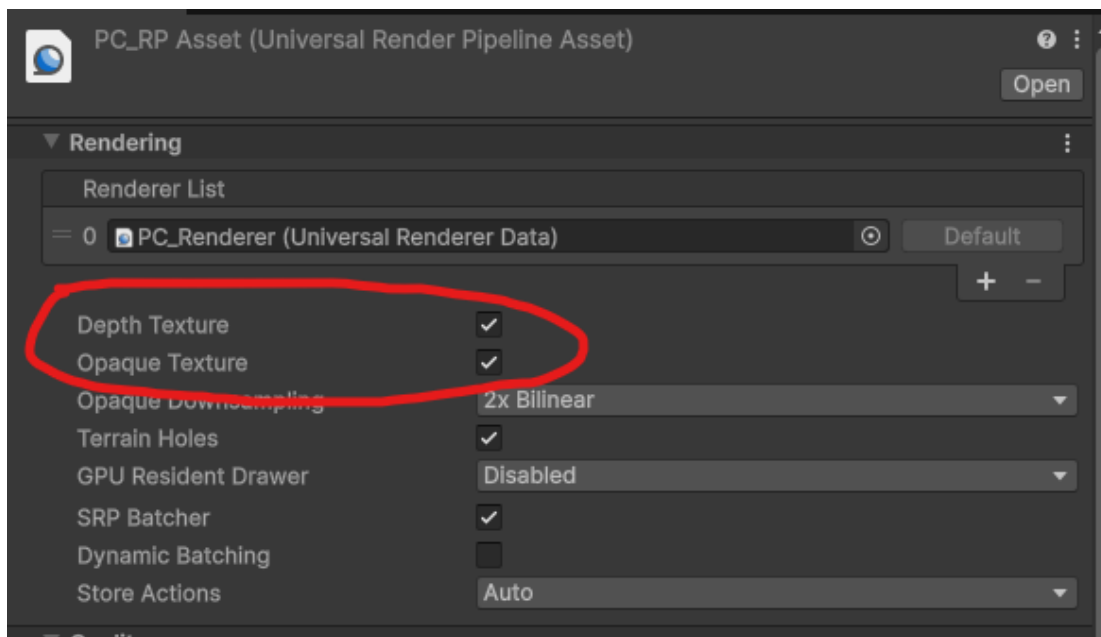
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Technical Architecture & Pipeline Overview

The StylizedFluids shader is a high-performance shading solution created specifically for stylized fluids inside Unity's Universal Render Pipeline. It combines gerstner wave vertex displacement with a multi-layered masked noise. The shader utilizes a world-space depth and parallax to add depth to the fluid. There is refraction as well if users decide to use a custom normal map (there is a sample one included).

Requirements

Ensure that you enable the **depth texture** and the **opaque texture** in your URP render asset. Otherwise, your transparency and depth will not work.



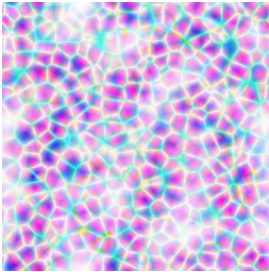
Also ensure your shader file source stays adjacent to your Noise.png texture. The FluidShaderGUI companion script executes folder path relative checks to locate this dependency automatically.

Main Features

- **Vertex Shader (Displacement):** Utilizes three blended octaves of Gerstner Wave mathematics to simulate variable fluid crests in world space.
- **Fragment Shader (Layering):** Samples up to 3 layers layered on top of each other (Surface, Subsurface, and Intersection Foam lines). This utilizes dedicated keywords to drop unnecessary calculations on a per-material basis.
- **Color-Space Mapping:** Blends colors in HSV space to maintain vibrant value and saturation gradients before converting back to an RGB color output. For each layer, the GUI has a Color and a Vector4 to control the colors. The Color variable is used for the UI. The Vector4 is created from the Color and used in the HSV blending.

Complete Property Block Reference Manual

Inspector Field Label	Internal HLSL Uniform Name	Data Struct Type	Functional Description
Flow Direction	_FlowDirection	Float (0.0, 6.28)	Controls which direction the fluid flows in.
Depth	_Depth	Float	Controls how deep the visible portion of the fluid is.
Wave Speed	_WaveSpeed	Float	Determines the speed of the Gerstner Waves.
LOD Fade Range	_LODFadeRange	Vector4	Controls the distance at which the patterns are visible.
Deep Color	_DeepColor / _DeepColorHSV	Color / Vector4	The color of the deepest parts of the fluid.
Shallow Color	_ShallowColor / _ShallowColorHSV	Color / Vector4	The color of the fluid near the surface.
Subsurface Color	_SubsurfaceColor / _SubsurfaceColorHSV	Color / Vector4	The color of the pattern below the surface of the fluid.
Surface Color	_SurfaceColor / _SurfaceColorHSV	Color / Vector4	The color of the pattern on the surface of the fluid.
Intersection Color	_IntersectionColor / _IntersectionColorHSV	Color / Vector4	The color of the pattern found at the intersecting geometry of the fluid.
Subsurface Cutoff	_SubsurfaceCutoff	Vector4	Determines the cutoff of the subsurface pattern
Surface Cutoff	_SurfaceCutoff	Vector4	Determines the cutoff of the surface pattern
Intersection Cutoff	_IntersectionCutoff	Vector4	Determines the cutoff of the intersection pattern.
Subsurface Speed	_SubsurfaceSpeed	Float	Controls the speed of the subsurface pattern.
Surface Speed	_SurfaceSpeed	Float	Controls the speed of the surface pattern.

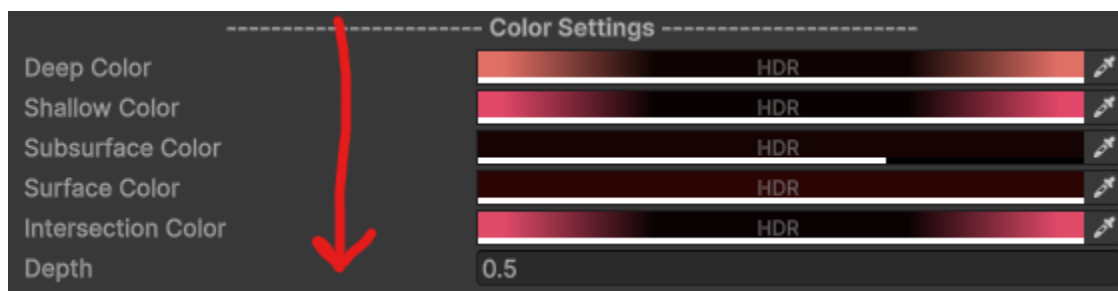
Inspector Field Label	Internal HLSL Uniform Name	Data Struct Type	Functional Description
Intersection Speed	_IntersectionSpeed	Float	Controls the speed of the intersection pattern.
Subsurface Scale	_SubsurfaceScale	Float	Controls the scale of the subsurface pattern.
Surface Scale	_SurfaceScale	Float	Controls the scale of the surface pattern.
Intersection Scale	_IntersectionScale	Float	Controls the speed of the intersection pattern.
Normal Strength	_NormalStrength	Float (0.0, 1.0)	Controls how strong the normal map is. Also influences the strength of refraction.
Normal Scale	_NormalScale	Float	Controls the scale of the normal texture.
Noise Texture	_NoiseTex	Texture2D (RGBA) 	Multi-channel noise map where X=Center Voronoi, Y=Edge Voronoi, Z=Domain Warped Noise, and W=Simplex Noise.
Surface Texture	_SurfaceTex	Texture2D (RGBA)	Optional Texture for the surface if the user wants to have their own pattern.
Subsurface Texture	_SubsurfaceTex	Texture2D (RGBA)	Optional Texture for the subsurface if the user wants to have their own pattern.
Normal Texture	_NormalTex	Texture2D (Normal)	Used for refraction and finer detail on the fresnel/lighting.
Wave Steepness	_WaveSteepness	Float (0.0 to 1.0)	Changes how much pinching occurs at the crests of the Gerstner Waves. Higher values produce sharper peaks.
Wavelength	_Wavelength	Float	Defines the wavelength of

Inspector Field Label	Internal HLSL Uniform Name	Data Struct Type	Functional Description
			the Gerstner Waves.
Surface Pattern	_SurfacePattern	Int	Controls which surface pattern shows.
Subsurface Pattern	_SubsurfacePattern	Int	Controls which subsurface pattern shows.
Intersection Pattern	_IntersectionPattern	Int	Controls which intersection pattern shows.
Recalculate Normals	_RecalculateNormals	Float (Toggle)	Toggles whether normals are recalculated due to vertex displacement from Gerstner Waves
World Space UVs	_WorldSpaceUVs	Float (Toggle)	Toggles whether the patterns are projected in UV space or World Space.
Line Density	_LineDensity	Float	Determines how many distinct lines are visible in the intersection outlines.
Parallax Amount	_Parallax	Float (0.0 to 0.1)	Controls how much parallax is present in the subsurface pattern.

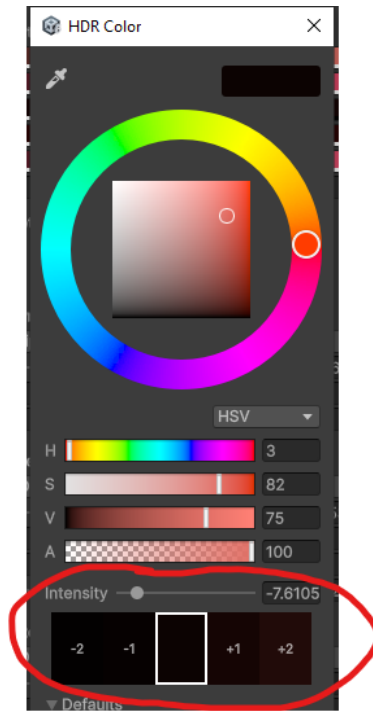
Tips on Efficient Usage

Color Selection

Due to the HDR colors and HSV blending, it can be tricky to pick the colors quickly. I found that selecting the bottom-most layer first and working your way to the surface has the best results.

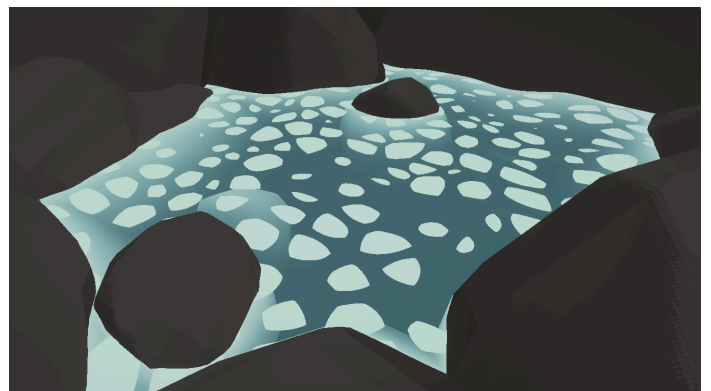
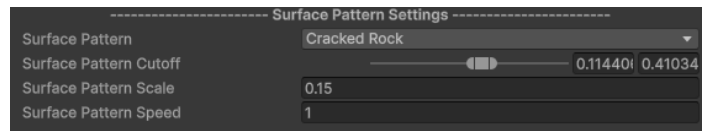
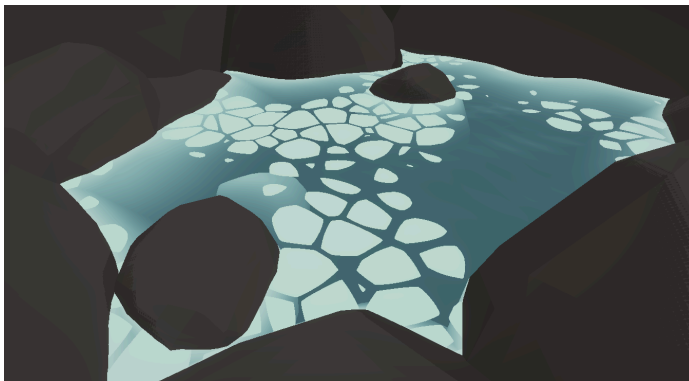
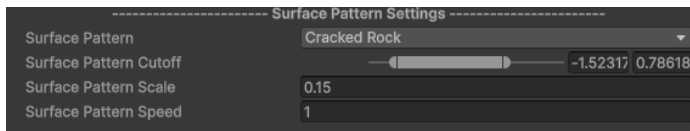


Additionally, the HDR makes even the smallest changes have a huge impact. Changing the intensity to negative values can help you get darker values. Positive intensities will cause a glow if you have bloom enabled on your post processing.

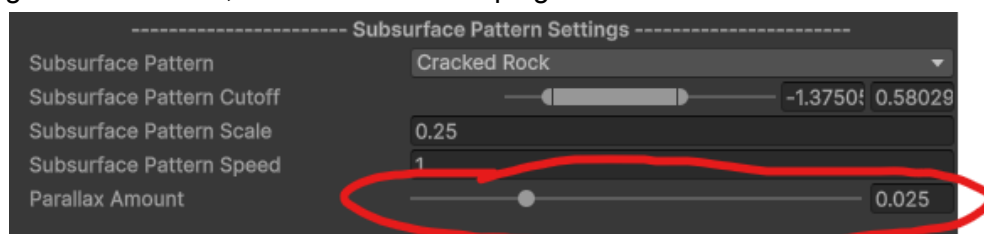


Pattern Selection

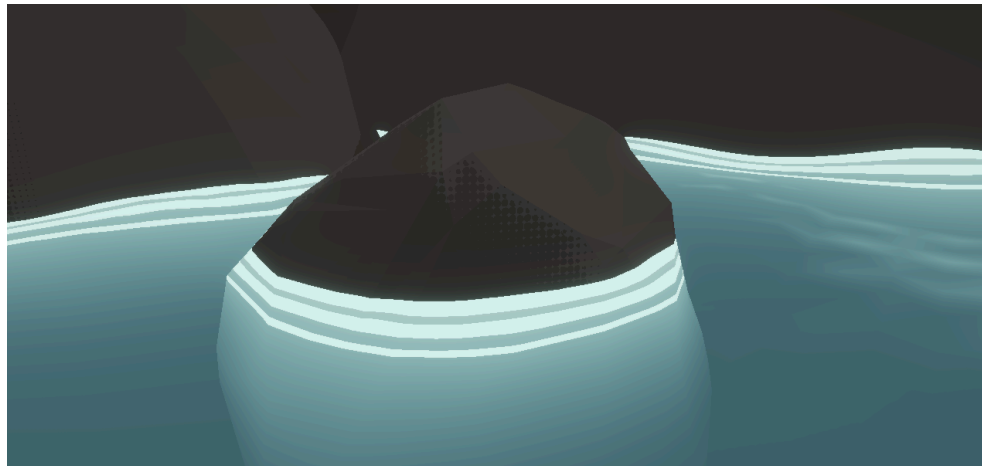
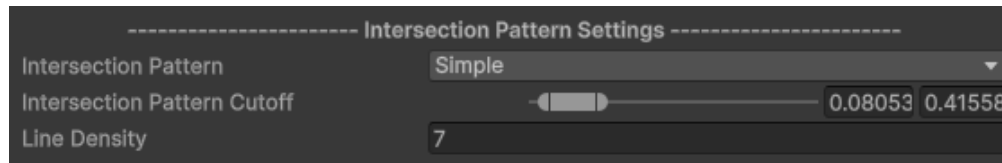
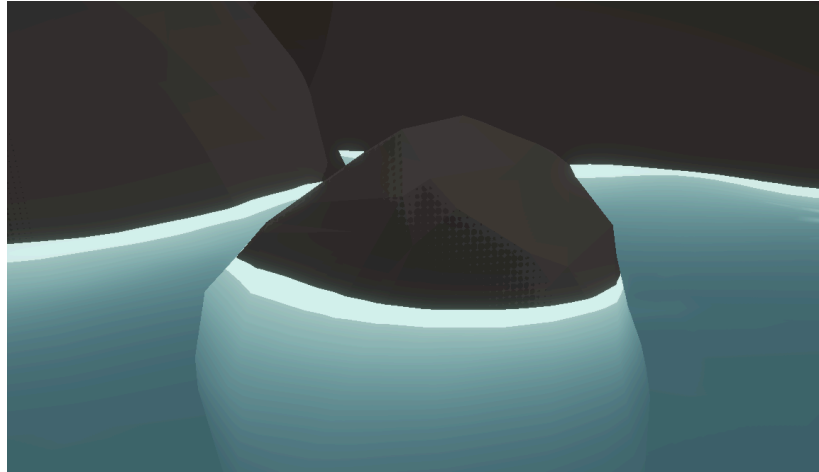
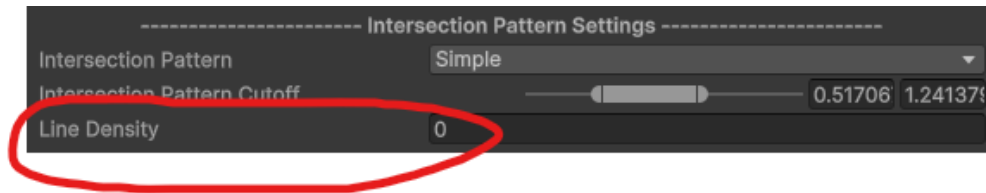
After selecting a pattern, you will notice a slider. This slider determines how your pattern will be “cutoff”. Smaller ranges will have a more evenly dispersed pattern, while larger ones will be more “patchy”.



The subsurface pattern section includes an extra parameter called “parallax amount”. This provides the subsurface pattern with parallax to add depth. One downside is that when “recalculate normals” are enabled and the wave heights are extreme, this can lead to warping.



For the intersection pattern section, you’ll see a parameter for line density.



This parameter controls how many intersections lines there can be. I would recommend mostly using this for the simple pattern type, but in some weird cases, I could get some cool results with the other types as well. You will need to mess with the cutoff for this as well.

LOD Fade Range

The LOD Fade Range helps keep the shader from looking rough in the distance. The first number determines when the patterns will begin to disappear. The second number determines when they will be completely gone.

Shader Keywords

The following pragmas were used in my shader to help keep the shader performant friendly while remaining as

versatile as it is.

Active Material Compilation Keyword	Conditional Processing Execution Behavior
_SURFACE_PATTERN_TYPE_FLOWMAP	Used if the flowmap pattern is chosen for the surface pattern. Note: This is the most expensive choice for the surface pattern.
_SURFACE_PATTERN_TYPE_TEXTURE	Used if a texture sample pattern is used for the surface pattern.
_SURFACE_PATTERN_TYPE_PATTERN	Used if a procedural pattern is chosen for the surface pattern.
_SUBSURFACE_PATTERN_TYPE	Differentiates between procedural patterns and texture sample related patterns for the subsurface patterns.
_INTERSECTION_PATTERN_TYPE	Differentiates between procedural patterns and texture sample related patterns for the intersection patterns.
_USE_NORMAL_MAP	Activates additional TBN surface detail projections, screen refraction math, and scene-depth offset check.
_WAVES	Enables Gerstner Wave calculations. If disabled, calculations are skipped.

Release Notes

Version 1.0.0

Additions

- Initial Release

Improvements

- Initial Release

Optimizations

- Initial Release

Notes

- Initial Release